

Maharashtra Public Service Commission Assessment

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Assistant Professor Botany Maharashtra Education Service Group A

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Assistant Professor Botany Maharashtra Education Service Group A

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Question Number : 1 Question Id : 4410092119317 Question Type : MCQ Option Shuffling : No

The major lipid-soluble antioxidant that terminates free radical chain reactions during membrane lipid peroxidation is:

Options :

1. Vitamin D
2. Vitamin A
3. Vitamin C
4. Vitamin E

Question Number : 2 Question Id : 4410092119320 Question Type : MCQ Option Shuffling : No

Which type of interaction mainly drives the folding of nonpolar regions in proteins?

Options :

1. Hydrophobic effect
2. Hydrogen bonding
3. Covalent bonding
4. Ionic bonding

Question Number : 3 Question Id : 4410092119330 Question Type : MCQ Option Shuffling : No

Which DNA helix has the widest diameter among the common forms?

Options :

1. Z-DNA
2. A-DNA
3. C-DNA
4. B-DNA

Question Number : 4 Question Id : 4410092119333 Question Type : MCQ Option Shuffling : No

Which process best describes how glucose is converted into pyruvate during carbohydrate metabolism in plant cells?

Options :

1. Pentose phosphate pathway
2. Tricarboxylic acid cycle
3. Glycolysis
4. Calvin cycle

Question Number : 5 Question Id : 4410092119324 Question Type : MCQ Option Shuffling : No

Which colligative property is used to determine the molar mass of an unknown solute in solution?

Options :

1. Osmometry
2. Polarity
3. Conductivity
4. Reactivity

Question Number : 6 Question Id : 4410092119326 Question Type : MCQ Option Shuffling : No

In a coupled reaction, what is the primary role of the molecule with a large negative free energy change?

Options :

1. To drive the endergonic process
2. To store genetic information
3. To absorb the energy of the system
4. To transport electrons

Question Number : 7 Question Id : 4410092119329 Question Type : MCQ Option Shuffling : No

What is the main advantage of the induced fit model over the lock and key model in enzyme catalysis?

Options :

1. Fixed structure of active site
2. Absolute rigidity of active site
3. Increased solubility of active site
4. Conformational adjustability of active site

Question Number : 8 Question Id : 4410092119335 Question Type : MCQ Option Shuffling : No

Which nucleotide is directly synthesized from inosine monophosphate (IMP) in plants?

Options :

1. Guanosine monophosphate
2. Cytidine monophosphate
3. Uridine monophosphate
4. Adenosine monophosphate

Question Number : 9 Question Id : 4410092119974 Question Type : MCQ Option Shuffling : No

Which post translational modification begins in the rough endoplasmic reticulum (RER) for most soluble secretory proteins?

Options :

1. N-linked glycosylation
2. O-linked glycosylation
3. Cleavage of mRNA
4. Assembly of clathrin coats

Question Number : 10 Question Id : 4410092119984 Question Type : MCQ Option Shuffling : No

Which of the following pairs is **CORRECTLY** matched with its role during mitosis?

Options :

1. Centromere — Site of ATP synthesis
2. Nucleolus — Chromosome segregation center
3. Centrosome — Site of DNA replication
4. Kinetochore — Attachment site for spindle microtubules

Question Number : 11 Question Id : 4410092119970 Question Type : MCQ Option Shuffling : No

Which statement about SNARE proteins involved in vesicle transport is **CORRECT**?

Options :

1. v-SNAREs are located on vesicles, whereas t-SNAREs are located on target membranes.
2. v-SNAREs and t-SNAREs are both present only on vesicle membranes.
3. SNARE proteins use NADH hydrolysis directly to drive membrane fusion.
4. SNARE proteins function only after membrane fusion is complete.

Question Number : 12 Question Id : 4410092119976 Question Type : MCQ Option Shuffling : No

Which of the following complexes is responsible for acidifying the vacuolar lumen by transporting protons?

Options :

1. ATP synthase
2. Vacuolar H⁺-ATPase
3. Cytochrome c oxidase
4. Na⁺/K⁺-ATPase

Question Number : 13 Question Id : 4410092119971 Question Type : MCQ Option Shuffling : No

A cell membrane suddenly becomes highly permeable to Na⁺ ions. What is the immediate effect on membrane potential?

Options :

1. Hyperpolarization
2. No change
3. Membrane dissolution
4. Depolarization

Question Number : 14 Question Id : 4410092119972 Question Type : MCQ Option Shuffling : No

Which of the following statements about the nucleus is **INCORRECT**?

Options :

1. The nucleolus is involved in ribosome assembly.
2. Nuclear pores allow regulated transport across the nuclear envelope.
3. The nuclear envelope consists of a double membrane.
4. DNA replication occurs within the nucleolus.

Question Number : 15 Question Id : 4410092119975 Question Type : MCQ Option Shuffling : No

A cell lacking functional peroxisomes will show impaired _____.

Options :

1. synthesis of ribosomal RNA
2. packaging of secretory proteins
3. detoxification of hydrogen peroxide
4. assembly of microtubules

Question Number : 16 Question Id : 4410092119983 Question Type : MCQ Option Shuffling : No

Compared with heterochromatin, euchromatin form generally shows _____.

Options :

1. lower rates of gene expression
2. greater accessibility to transcription factors
3. higher levels of chromatin condensation
4. increased centromeric DNA content

Question Number : 17 Question Id : 4410092119991 Question Type : MCQ Option Shuffling : No

Which event occurs during replication of linear eukaryotic chromosomes but is generally absent in replication of circular prokaryotic chromosomes?

Options :

1. Leading-strand synthesis
2. RNA primer formation
3. Telomere elongation
4. Bidirectional replication

Question Number : 18 Question Id : 4410092119992 Question Type : MCQ Option Shuffling : No

Which cellular mechanism allows replication to continue and prevents fork collapse when UV-induced DNA damage blocks the progression of replication forks?

Options :

1. Non-homologous end joining
2. Lesion bypass
3. Mismatch repair
4. Base excision repair

Question Number : 19 Question Id : 4410092119996 Question Type : MCQ Option Shuffling : No

Recruitment of histone deacetylases (HDACs) by a transcription repressor generally leads to_____.

Options :

1. increased chromatin accessibility
2. enhanced RNA processing
3. increased translation efficiency
4. reduced gene transcription

Question Number : 20 Question Id : 4410092120000 Question Type : MCQ Option Shuffling : No

Alternative splicing allows a single gene to produce multiple proteins by _____.

Options :

1. using different promoters only
2. joining exons in different combinations
3. replicating DNA repeatedly
4. increasing ribosome number

Question Number : 21 Question Id : 4410092119997 Question Type : MCQ Option Shuffling : No

An mRNA undergoes normal splicing and polyadenylation but remains highly unstable. Which modification is most likely missing?

Options :

1. 5' cap
2. Poly(A) tail
3. Intron removal
4. Promoter sequence

Question Number : 22 Question Id : 4410092119999 Question Type : MCQ Option Shuffling : No

The failure of RNA processing interferes with protein synthesis because:

Options :

1. Histones cannot bind DNA
2. DNA synthesis stops completely
3. Chromosomes cannot align
4. Mature mRNA production is impaired

Question Number : 23 Question Id : 4410092120007 Question Type : MCQ Option Shuffling : No

Which of the following steps is most closely associated with translational proofreading during protein synthesis?

Options :

1. Aminoacyl-tRNA accommodation
2. RNA interference
3. Alternative splicing
4. Ribosome recycling

Question Number : 24 Question Id : 4410092120008 Question Type : MCQ Option Shuffling : No

In bacteriophage λ , establishment of the lysogenic cycle is associated with the synthesis of _____.

Options :

1. Cro protein
2. CI repressor protein
3. RNA polymerase
4. Sigma factor

Question Number : 25 Question Id : 4410092120012 Question Type : MCQ Option Shuffling : No

The first essential step in successful pathogen entry into a host cell is _____.

Options :

1. Protein synthesis by the pathogen
2. Attachment to specific host-cell receptors
3. Host-cell lysis
4. Replication of pathogen genome

Question Number : 26 Question Id : 4410092120023 Question Type : MCQ Option Shuffling : No

Which cellular process allows bacteria to alter their direction during chemotaxis?

Options :

1. Histone modification
2. Change in flagellar rotation
3. DNA recombination
4. Ribosomal assembly

Question Number : 27 Question Id : 4410092120024 Question Type : MCQ Option Shuffling : No

Which growth factor primarily regulates erythrocyte production during hematopoiesis?

Options :

1. Thrombopoietin
2. Erythropoietin
3. Interleukin-2
4. Transforming growth factor- β

Question Number : 28 Question Id : 4410092120030 Question Type : MCQ Option Shuffling : No

Complete the analogy.

p53 : DNA damage checkpoint :: Rb : _____

Options :

1. Regulation of G1-S transition
2. DNA replication initiation
3. Chromosome segregation
4. Telomere elongation

Question Number : 29 Question Id : 4410092112940 Question Type : MCQ Option Shuffling : No

What is the primary mechanism by which viral oncogenes promote cancer development in host cells?

Options :

1. They force uncontrolled expression of growth proteins
2. They completely block viral replication inside cells
3. They directly degrade host DNA strands rapidly
4. They inhibit all metabolic pathways in host cells

Question Number : 30 Question Id : 4410092120015 Question Type : MCQ Option Shuffling : No

A viral mutation permits normal host-cell receptor binding but abolishes activation of the viral fusion protein. Which stage of the infection process is directly blocked?

Options :

1. Host-cell receptor recognition
2. Viral genome replication
3. Viral transcription
4. Fusion of viral and cellular membranes

Question Number : 31 Question Id : 4410092120020 Question Type : MCQ Option Shuffling : No

In a GPCR signaling pathway, a $G\alpha$ subunit remains GTP-bound due to a mutation affecting its intrinsic GTPase activity. This will result in_____.

Options :

1. Persistent stimulation of downstream effector proteins
2. Failure of receptor activation by ligand
3. Inability of all the receptor to bind G proteins
4. Reduced synthesis of signaling molecules

Question Number : 32 Question Id : 4410092120028 Question Type : MCQ Option Shuffling : No

What will be the outcome of a chromosomal translocation positioning a proto-oncogene under the control of a highly active promoter?

Options :

1. Reduced gene expression
2. Overexpression of proto-oncogene
3. Enhanced DNA repair activity
4. Permanent gene silencing

Question Number : 33 Question Id : 4410092119670 Question Type : MCQ Option Shuffling : No

What is the term for a gene from one species that is artificially inserted into the genome of another plant species to confer a new trait, such as herbicide resistance?

Options :

1. Transgene
2. Allele
3. Marker gene
4. Operon

Question Number : 34 Question Id : 4410092119679 Question Type : MCQ Option Shuffling : No

Which specific cells within the embryo sac are responsible for directing the pollen tube during the final stages of its guidance?

Options :

1. Antipodal
2. Synergid
3. Central
4. Egg

Question Number : 35 Question Id : 4410092119683 Question Type : MCQ Option Shuffling : No

In Arabidopsis root development, what is the primary role of the SHORT-ROOT (SHR) and SCARECROW (SCR) genes?

Options :

1. Root tissue patterning
2. Root exudate secretion
3. Root bacterial association
4. Root water uptake

Question Number : 36 Question Id : 4410092119684 Question Type : MCQ Option Shuffling : No

During the floral transition in Arabidopsis, the shoot apical meristem first converts into _____ .

Options :

1. Primary inflorescence meristem
2. Axillary meristem
3. Cauline leaf primordium
4. Vegetative meristem

Question Number : 37 Question Id : 4410092119671 Question Type : MCQ Option Shuffling : No

In most angiosperm shoot apical meristems, what is the characteristic pattern of cell division in the tunica?

Options :

1. Random anticlinal and periclinal cell divisions
2. Equal anticlinal and periclinal cell divisions
3. Periclinal cell divisions only
4. Predominantly anticlinal cell divisions

Question Number : 38 Question Id : 4410092119682 Question Type : MCQ Option Shuffling : No

During early embryogenesis in plants, which protein's polar localization is primarily responsible for directing auxin accumulation that leads to shoot apical meristem (SAM) formation?

Options :

1. AGP proteins
2. SCR proteins
3. RAD proteins
4. PIN proteins

Question Number : 39 Question Id : 4410092119685 Question Type : MCQ Option Shuffling : No

The cloning of the mutated **RADIALIS (RAD)** gene in *Antirrhinum majus* was used to understand _____.

Options :

1. Floral asymmetry specification
2. Floral organ identity determination
3. Meristem identity transition control
4. Sepal and petal development regulation

Question Number : 40 Question Id : 4410092119688 Question Type : MCQ Option Shuffling : No

Why does oxidative metabolism contribute to programmed cell death during leaf aging?

Options :

1. Decrease due to high antioxidant activity
2. Increase as antioxidant defences decline
3. Remain relatively constant throughout senescence
4. Stop completely during the aging process

Question Number : 41 Question Id : 4410092119693 Question Type : MCQ Option Shuffling : No

Which process is responsible for ATP synthesis in chloroplasts during photosynthesis?

Options :

1. Decarboxylation
2. Fermentation
3. Photophosphorylation
4. Glycolysis

Question Number : 42 Question Id : 4410092119694 Question Type : MCQ Option Shuffling : No

Which molecule is transported from the chloroplast to the peroxisome at the beginning of photorespiration?

Options :

1. Serine
2. Glutamate
3. Glycolate
4. Glycerate

Question Number : 43 Question Id : 4410092119691 Question Type : MCQ Option Shuffling : No

Which enzyme is involved in the initial nighttime carbon dioxide fixation in the CAM (Crassulacean Acid Metabolism) photosynthesis?

Options :

1. Nitrate reductase
2. Phosphoenolpyruvate (PEP) carboxylase
3. Catalase
4. Ribulose-1,5-bisphosphate (RuBP) oxygenase

Question Number : 44 Question Id : 4410092119709 Question Type : MCQ Option Shuffling : No

Which of the following is a nitrogen-containing secondary metabolite found in plants?

Options :

1. Glucose
2. Starch
3. Caffeine
4. Cellulose

Question Number : 45 Question Id : 4410092119698 Question Type : MCQ Option Shuffling : No

Which biochemical process directly reduces the concentration of active abscisic acid (ABA) concentration in leaf tissue after rewatering of drought-stressed plants?

Options :

1. Redistribution of ABA from leaves to other plant organs via phloem transport
2. Oxidative catabolism of ABA to phaseic acid in leaf tissues
3. Reduced ABA biosynthesis in vascular tissues after water potential recovery
4. Enhanced conjugation of ABA into inactive ABA-glucose ester forms

Question Number : 46 Question Id : 4410092119699 Question Type : MCQ Option Shuffling : No

Which vascular tissue transports gibberellins synthesized in shoot leaves downward toward the root system?

Options :

1. Xylem tissue
2. Cortical tissue
3. Epidermal tissue
4. Phloem tissue

Question Number : 47 Question Id : 4410092119702 Question Type : MCQ Option Shuffling : No

Based on experiments with CRY2 mutants in *Arabidopsis*, what primary role do cryptochromes serve in plant sensory photobiology?

Options :

1. Hormones that directly regulate auxin transport
2. Enzymes involved in chlorophyll degradation
3. Red-light receptors controlling seed germination only
4. Blue-light photoreceptors involved in photoperiod regulation

Question Number : 48 Question Id : 4410092119708 Question Type : MCQ Option Shuffling : No

Which process describes phloem loading in plants?

Options :

1. Diffusion of gases occurring in leaf air spaces
2. Movement of water from xylem into nearby phloem tissues
3. Transfer of photosynthates from source cells into sieve elements
4. Breakdown of sugars occurring within sink tissues

Question Number : 49 Question Id : 4410092119711 Question Type : MCQ Option Shuffling : No

Which term describes a trait that is expressed in the F1 generation when two different alleles are present?

Options :

1. Dominant trait
2. Intermediate trait
3. Mutant trait
4. Recessive trait

Question Number : 50 Question Id : 4410092119714 Question Type : MCQ Option Shuffling : No

What is the main goal of performing a complementation test in genetics?

Options :

1. To measure rate of mutations in a population over time
2. To study the mechanism of DNA replication in cells
3. To determine mutations are in the same gene or different gene
4. To estimate the number of chromosomes in a genome

Question Number : 51 Question Id : 4410092119713 Question Type : MCQ Option Shuffling : No

What characteristic led to the original proposal of the term 'pseudoalleles' for certain mutant elements?

Options :

1. Produced identical mutant phenotypes
2. Occupied exactly the same nucleotide position in DNA
3. Behaved as alleles but could recombine with each other
4. Showed complete linkage without recombination

Question Number : 52 Question Id : 4410092119718 Question Type : MCQ Option Shuffling : No

Crossing over can break up associations between genes that are:

Options :

1. Present only in mitochondrial genome of the cell
2. Located close together on the same chromosome
3. Found in organisms of different species groups
4. Located on entirely separate chromosomes

Question Number : 53 Question Id : 4410092119727 Question Type : MCQ Option Shuffling : No

A pedigree chart shows that two unaffected parents have a son and a daughter who both express a genetic trait. The trait is not observed in the parental generation and is equally present in both sexes. What is the mode of inheritance for this trait?

Options :

1. X-linked dominant
2. Autosomal recessive
3. Y-linked
4. Autosomal dominant

Question Number : 54 Question Id : 4410092119717 Question Type : MCQ Option Shuffling : No

In polydactyly, some affected individuals have fully functional extra digits, whereas others have only a small skin tag. This difference is an example of:

Options :

1. Variable expressivity
2. Multiple alleles
3. Incomplete dominance
4. Codominance

Question Number : 55 Question Id : 4410092119722 Question Type : MCQ Option Shuffling : No

When developing a mapping population from a cross between two genetically distant parent plants, which outcome regarding genetic variation is to be observed?

Options :

1. Complete homozygosity
2. No segregation
3. Increased polymorphism
4. Reduced polymorphism

Question Number : 56 Question Id : 4410092119723 Question Type : MCQ Option Shuffling : No

A researcher studies a chloroplast-encoded trait in a plant species. The trait is transmitted from affected females to all progeny, whereas affected males never transmit the trait. Which mechanism best explains this pattern?

Options :

1. Independent inheritance through nuclear chromosomes only
2. Sex-linked inheritance through maternal transmission only
3. Maternal inheritance through exclusion of pollen chloroplasts
4. Cytoplasmic segregation through meiotic organelle sorting

Question Number : 57 Question Id : 4410092119733 Question Type : MCQ Option Shuffling : No

In conditional mutants, which term specifically describes the environmental state that permits the expression of the mutant phenotype?

Options :

1. Restrictive condition
2. Recessive condition
3. Dominant condition
4. Neutral condition

Question Number : 58 Question Id : 4410092119739 Question Type : MCQ Option Shuffling : No

During the production of transgenic mice by pronuclear microinjection, which mechanism describes how injected linear DNA typically integrates into the host chromosome?

Options :

1. Homologous recombination
2. Crossing over during meiosis
3. Nonhomologous recombination
4. Independent assortment

Question Number : 59 Question Id : 4410092119515 Question Type : MCQ Option Shuffling : No

Which one of the following is the lowest rank in the taxonomic hierarchy?

Options :

1. Species
2. Genus
3. Family
4. Order

Question Number : 60 Question Id : 4410092119527 Question Type : MCQ Option Shuffling : No

Which of the following feature pairs distinguishes a manual from a flora in taxonomic literature?

Options :

1. Manual: global plants, Flora: cultivated plants.
2. Manual: practical identification keys, Flora: describes species in a region.
3. Manual: detailed species descriptions, Flora: provides only illustrations.
4. Manual: lists only rare species, Flora: includes all species.

Question Number : 61 Question Id : 4410092119529 Question Type : MCQ Option Shuffling : No

What is a practical limitation in the identification of a cryptic species in biosystematics?

Options :

1. Morphological traits do not reflect genetic differences.
2. Absence of type specimens in herbaria.
3. DNA barcoding always fail to detect genetic variation.
4. Lack of evolutionary history data.

Question Number : 62 Question Id : 4410092119533 Question Type : MCQ Option Shuffling : No

A species is classified as 'Extinct in the Wild (EW)' by the IUCN if _____.

Options :

1. no living individuals remain on Earth
2. it survives only outside its native range
3. it has a stable but small wild population
4. it is threatened but still present in the wild

Question Number : 63 Question Id : 4410092119516 Question Type : MCQ Option Shuffling : No

According to the principle of legitimacy in botanical nomenclature, a name is considered illegitimate if _____.

Options :

1. it is published after 1953
2. it is used in horticultural trade
3. it contains a typographical error
4. it was superfluous when published

Question Number : 64 Question Id : 4410092119519 Question Type : MCQ Option Shuffling : No

Which of the following statements clearly presents a peculiar feature of Volvox reproduction?

Options :

1. All cells are reproductive.
2. Colony reproduces by fragmentation.
3. Gonidia form daughter colonies.
4. Somatic cells form spores.

Question Number : 65 Question Id : 4410092119522 Question Type : MCQ Option Shuffling : No

Why are reproductive traits of greater taxonomic importance as compared to vegetative traits in plant classification systems?

Options :

1. They reflect evolutionary relationships effectively.
2. They show similar complexity in different plant groups.
3. They evolve more rapidly than vegetative structures.
4. They are easily affected by the environmental factors.

Question Number : 66 Question Id : 4410092119532 Question Type : MCQ Option Shuffling : No

Which of the following criteria is considered for designating a region as a biodiversity hotspot according to Norman Myers?

Options :

1. High level of plant endemism and extensive habitat loss.
2. Overall high species richness and large forest cover.
3. High level of animal endemism and low human impact.
4. Large area of undisturbed habitat with species richness.

Question Number : 67 Question Id : 4410092119538 Question Type : MCQ Option Shuffling : No

Which of the following statements defines habitat and niche CORRECTLY?

Options :

1. Habitat and niche both means the space occupied.
2. Niche is the range of behaviours, habitat is space occupied.
3. Habitat is the physical space, niche is the functional role.
4. Habitat and niche both mean a range of behaviours.

Question Number : 68 Question Id : 4410092119542 Question Type : MCQ Option Shuffling : No

Which one of the following parameters measures the reproductive output of a population?

Options :

1. Density
2. Natality
3. Dispersion
4. Mortality

Question Number : 69 Question Id : 4410092119537 Question Type : MCQ Option Shuffling : No

What is the relationship between ecological amplitude and habitat range?

Options :

1. Wide amplitude enables broad habitat range.
2. All species have narrow amplitude regardless of habitat.
3. Wide habitat range reduces ecological amplitude.
4. Ecological amplitude is unrelated to habitat range.

Question Number : 70 Question Id : 4410092119553 Question Type : MCQ Option Shuffling : No

Which biogeographical zone in India is characterized by high rainfall, dense evergreen forests, and high endemism?

Options :

1. Thar Desert
2. Deccan Plateau
3. Gangetic Plain
4. Western Ghats

Question Number : 71 Question Id : 4410092119539 Question Type : MCQ Option Shuffling : No

How does niche overlap influence the potential for hybridization between two closely related plant species?

Options :

1. It increases hybridization due to shared pollinators and flowering times.
2. It reduces hybridization by enforcing spatial segregation.
3. It has no effect on hybridization as species remain isolated.
4. It eliminates hybridization due to increased competition.

Question Number : 72 Question Id : 4410092119544 Question Type : MCQ Option Shuffling : No

A botanist studies the plant population of a region using the Leslie matrix. What characteristic of the population does this approach describe?

Options :

1. Mating occurs randomly among all age groups in a population.
2. Environmental conditions affect survival of all plant species.
3. In a population, all age classes have equal survival rates.
4. Transitions between age classes occur at fixed time intervals.

Question Number : 73 Question Id : 4410092119547 Question Type : MCQ Option Shuffling : No

In a disturbed area, the dominance of one species increases, but the total number of species remains constant. What is the impact on Simpson's diversity index ($1 - D$)?

Options :

1. Simpson's index value increases
2. Simpson's index remains constant
3. Simpson's index value decreases
4. Simpson's index value becomes 0

Question Number : 74 Question Id : 4410092119555 Question Type : MCQ Option Shuffling : No

A conservation team uses the Shannon-Wiener Index to monitor biodiversity in a restored wetland. What does a significant decline in the index value indicate?

Options :

1. Increase in dominance by rare species
2. Reduction in species richness and evenness
3. Increase in ecosystem productivity
4. Alteration in structural complexity of habitat

Question Number : 75 Question Id : 4410092119558 Question Type : MCQ Option Shuffling : No

According to Darwin's theory, the individuals with less favourable traits _____

Options :

1. are less likely to survive and reproduce.
2. evolve faster than other individuals.
3. migrate to favorable environments.
4. are more likely to survive and reproduce.

Question Number : 76 Question Id : 4410092119561 Question Type : MCQ Option Shuffling : No

According to Haldane's theory, which of the following energy sources synthesized organic molecules on early Earth?

Options :

1. Lightning and ultraviolet radiation
2. Ionizing radiation
3. Chemical energy from respiration
4. Heat from animal metabolism

Question Number : 77 Question Id : 4410092119569 Question Type : MCQ Option Shuffling : No

What is the main theme of the neutral theory of molecular evolution?

Options :

1. Selectively neutral genetic mutations are fixed by genetic drift.
2. Disadvantageous mutations are removed by natural selection.
3. All mutations confer an advantage in changing environments.
4. Beneficial mutations drive molecular evolution.

Question Number : 78 Question Id : 4410092119571 Question Type : MCQ Option Shuffling : No

Which of the following processes is responsible for the emergence of novel proteins with unique functions in plant genomes?

Options :

1. Point mutations
2. Gene loss through deletion
3. Intron rearrangements
4. Exon shuffling

Question Number : 79 Question Id : 4410092119563 Question Type : MCQ Option Shuffling : No

Which of the following hypotheses explains the origin of genetic information storage in the first prokaryotic cells?

Options :

1. Self-replicating RNA molecules acted as the initial genetic system
2. Protein-based inheritance preceded nucleic acid systems
3. Horizontal gene transfer established initial genetic systems
4. Hybrid RNA-DNA molecules can be considered initial genetic material

Question Number : 80 Question Id : 4410092119573 Question Type : MCQ Option Shuffling : No

Why is random genetic drift more pronounced in small populations compared with large populations?

Options :

1. Genetic drift is counteracted by mutation rates in small populations.
2. Natural selection is completely absent in small populations.
3. Chance fluctuation impacts allele frequencies in a small population.
4. Gene flow increases the effect of drift in small populations.

Question Number : 81 Question Id : 4410092119574 Question Type : MCQ Option Shuffling : No

A plant species colonized a new island. The plant population showed a reduced genetic diversity and unique traits after several generations. Which of the following processes led to speciation in this case?

Options :

1. Bottleneck effect from environmental catastrophe.
2. Hybridization with endemic species in the area.
3. Founder effect followed by genetic divergence.
4. Gene flow maintaining genetic similarity in plants.

Question Number : 82 Question Id : 4410092119576 Question Type : MCQ Option Shuffling : No

How does sexual selection affect genetic drift in small, isolated plant populations?

Options :

1. Sexual selection completely removes effects of genetic drift
2. Sexual selection and genetic drift act independently of population size
3. Sexual selection may amplify random allele frequency changes
4. Sexual selection increases genetic diversity in small populations

Question Number : 83 Question Id : 4410092119838 Question Type : MCQ Option Shuffling : No

Which plant hormone combination in a tissue culture medium typically induces the formation of an unorganized mass of dividing cells (callus)?

Options :

1. High cytokinin and low auxin
2. High auxin and low cytokinin
3. Balanced levels of both auxin and cytokinin
4. High gibberellin and abscisic acid

Question Number : 84 Question Id : 4410092119843 Question Type : MCQ Option Shuffling : No

Which of the following global health issues is primarily targeted by the development of the transgenic crop known as 'Golden Rice'?

Options :

1. Iron deficiency anemia
2. Vitamin A deficiency
3. Protein-energy malnutrition
4. Scurvy

Question Number : 85 Question Id : 4410092119847 Question Type : MCQ Option Shuffling : No

Plant-based transient expression systems (magniffection) are used to rapidly produce recombinant viral vectors for human gene therapy. Which of the following viral backbone is typically utilized to ensure massive transcript amplification?

Options :

1. Cauliflower mosaic virus (CaMV)
2. Tobacco mosaic virus (TMV)
3. Tomato spotted wilt virus (TSWV)
4. Potato spindle tuber viroid (PSTVd)

Question Number : 86 Question Id : 4410092119849 Question Type : MCQ Option Shuffling : No

In Marker-Assisted Selection (MAS) for a recessive trait, what is the primary advantage of using co-dominant markers like Simple Sequence Repeats (SSRs) over dominant markers like RAPDs?

Options :

1. SSRs allow the differentiation of heterozygous carriers from homozygous resistant plants
2. SSRs do not require any prior sequence knowledge for primer design
3. SSRs generate a much higher number of polymorphic bands per PCR reaction
4. SSRs are exclusively located within the coding regions of target genes

Question Number : 87 Question Id : 4410092119834 Question Type : MCQ Option Shuffling : No

During the industrial production of glutamic acid by *Corynebacterium glutamicum*, which physiological manipulation is essential to trigger the efflux of the amino acid from the cell?

Options :

1. Increasing extracellular glucose
2. Inducing biotin deficiency
3. Supplementing with high levels of penicillin
4. Lowering the aeration rate

Question Number : 88 Question Id : 4410092119850 Question Type : MCQ Option Shuffling : No

During Marker-Assisted Backcrossing (MABC), why is the use of flanking markers tightly linked to the target gene critical during the "background selection" phase?

Options :

1. To identify off-target mutations caused by the initial transformation event
2. To minimize linkage drag by selecting rare recombinants close to the target locus
3. To ensure the target gene exhibits epistatic dominance over background genes
4. To increase the overall recombination rate across the recurrent genome

Question Number : 89 Question Id : 4410092119851 Question Type : MCQ Option Shuffling : No

In the microbial bioremediation of soil heavily contaminated with hydrophobic polycyclic aromatic hydrocarbons (PAHs), what is the primary rate-limiting step overcome by engineered *Pseudomonas* strains?

Options :

1. High oxygen toxicity; overcome by switching to strict anaerobic respiration
2. Low bioavailability; overcome by secreting rhamnolipid biosurfactants
3. Rapid PAH volatilization; overcome by forming protective extracellular biofilms
4. Nitrogen depletion; overcome by concurrent nitrogen fixation

Question Number : 90 Question Id : 4410092119852 Question Type : MCQ Option Shuffling : No

Transgenic plants engineered for phytoremediation often express bacterial *merA* and *merB* genes. What is the precise mechanism by which these plants remediate methylmercury?

Options :

1. They sequester methylmercury into root vacuoles via phytochelatins
2. They convert methylmercury to ionic mercury, which precipitates solidly in the roots
3. They cleave methylmercury and reduce it to elemental mercury, which is volatilized
4. They exude organic acids to bind and lock methylmercury in the rhizosphere

Question Number : 91 Question Id : 4410092119857 Question Type : MCQ Option Shuffling : No

Which of the following is commonly used as an initial step for isolating nucleic acid sequences from plant cells?

Options :

1. Phenol treatment
2. Mechanical tissue disruption
3. Heat shock
4. Protein precipitation

Question Number : 92 Question Id : 4410092119859 Question Type : MCQ Option Shuffling : No

Which feature distinguishes cosmid vectors from standard plasmid vectors in recombinant DNA techniques?

Options :

1. Cosmid vectors always carry antibiotic resistance genes for screening
2. Cosmid vectors contain *cos* sites enabling efficient DNA packaging
3. Cosmid vectors are always small and circular in structure
4. Cosmid vectors replicate only in eukaryotic cells and not in prokaryotic cells

Question Number : 93 Question Id : 4410092119858 Question Type : MCQ Option Shuffling : No

During the synthesis of double-stranded cDNA for library construction, what is the critical function of RNase H in the second-strand synthesis step?

Options :

1. It catalyzes the addition of poly-A tail sequences to the ends of mature messenger transcripts
2. It initiates the targeted degradation of the DNA strand within the transient hybrid molecule.
3. It creates nicks in the RNA strand of the hybrid to provide priming sites for DNA Polymerase.
4. It inhibits the formation of complex secondary structures within the single-stranded template.

Question Number : 94 Question Id : 4410092119864 Question Type : MCQ Option Shuffling : No

Electron spin resonance (ESR) spectroscopy uniquely detects chemical species that possess which of the following properties?

Options :

1. Unpaired electrons
2. High molecular weights
3. Net negative charges
4. Intrinsic fluorescence

Question Number : 95 Question Id : 4410092119877 Question Type : MCQ Option Shuffling : No

According to Abbe's equation for light microscopy, the resolving power of an objective lens is maximized by:

Options :

1. Increasing the wavelength of the light and decreasing numerical aperture
2. Decreasing the wavelength of the light and increasing numerical aperture
3. Using a condenser with a lower refractive index than air
4. Closing the iris diaphragm to its absolute minimum aperture

Question Number : 96 Question Id : 4410092119860 Question Type : MCQ Option Shuffling : No

To clone a large genomic DNA fragment (greater than 150 kb), which of the following vector systems would be most suitable for ensuring stability and propagation of such inserts in *Escherichia coli*?

Options :

1. Bacterial Artificial Chromosome (BAC) vector
2. Cosmid vector
3. Plasmid vector
4. Yeast Artificial Chromosome (YAC) vector

Question Number : 97 Question Id : 4410092119867 Question Type : MCQ Option Shuffling : No

If a Poisson model is used for the number of fungal infections per plant stem, which key assumption must be met?

Options :

1. The mean is always larger than the variance
2. The probability of infection increases with each event
3. Each infection occurs independently of others
4. There is a fixed number of plant stems observed

Question Number : 98 Question Id : 4410092119869 Question Type : MCQ Option Shuffling : No

If plant growth data shows consistent deviations across all replicates despite identical protocols, which statistical error is present and what is its main implication?

Options :

1. Observer error leading to accidental recording mistakes
2. Systematic error causing biased measurements
3. Sampling error leading to unrepresentative data
4. Random error causing unpredictable variation

Question Number : 99 Question Id : 4410092119871 Question Type : MCQ Option Shuffling : No

How many degrees of freedom are used in a Chi-square (X^2) goodness-of-fit test for an expected 9:3:3:1 Mendelian ratio?

Options :

1. 1
2. 3
3. 4
4. 15

Question Number : 100 Question Id : 4410092119873 Question Type : MCQ Option Shuffling : No

When utilizing induced micro-mutations for the genetic improvement of quantitative traits in crops, which statistical parameter is most critical for selecting promising lines in the M2 and M3 generations?

Options :

1. Shift in the theoretical segregation ratio of single genes
2. Increase in the genetic variance relative to the untreated control
3. Complete elimination of environmental variance
4. The exact nucleotide substitution rate in the coding region